

POTHUGANTI SIVA MEHAR GANATEJU

SYSTEMS ENGINEER | AI, ROBOTICS & EMBEDDED SYSTEMS

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EDUCATION

Vignan's Foundation for Science, Technology & Research (VFSTR)

B.Tech, Electronics & Communication Engineering — Minor: AI & ML | CGPA: 7.94

Hyderabad, India

Expected 2028

TECHNICAL COMPETENCIES

Systems Architecture: End-to-end system design, cross-subsystem integration, modular pipeline architecture, hardware/software co-design

Performance Engineering: Cache-aware data structures, throughput optimization, bitwise/low-level engineering, Linux performance profiling

Middleware & Integration: ROS2, PX4, MAVROS2, AirSim, Jetson, Firebase, Supabase, REST/API integration, SITL/HITL validation

Security Implementation: Zero-trust architecture, server-authoritative validation, JWT auth, environment-secured configuration, credential hardening

Languages & Frameworks: Python, C, C++, Java, JavaScript, Kotlin | PyTorch, TensorFlow, OpenCV, React, Next.js, Node.js, Express, Flask, Flutter

Platforms & Tooling: Git, PostgreSQL, MongoDB, MATLAB, Fusion 360, Arduino IDE, Custom PCB Design

SYSTEMS ARCHITECTURE & ENGINEERING PROJECTS

Binary Structural Fingerprint Indexing (BSFI)

June 2026

C++, Python, Bitwise Engineering — Deterministic Indexing Architecture

- Architected a deterministic structural coordinate indexing system using six binary descriptors, compressing 32-bit integers into packed 28-bit fingerprints and reducing candidate search space by 15,800× across 6,362,620 financial transaction records.
- Engineered a hardware-friendly C++ execution pipeline sustaining 40.6M fingerprints/sec with ~150× fewer CPU cache misses than linear scanning, validated through Linux performance counters.
- Designed a dual-coordinate retrieval subsystem (Primary Structural Fingerprint + Residual Null-Space coordinate), improving Top-10 retrieval precision from 49.97% to 72.13% through residual ranking.
- Achieved 98.45% hidden-fraud recovery coverage (7,101 of 7,213 records) at production scale, with zero retrieval drift confirmed across all validation queries.

Consensus Mesh

March 2026

Flutter, Node.js, MongoDB, JWT — Zero-Trust Distributed Verification Platform

- Architected a zero-trust, server-authoritative attendance verification system, eliminating all client-side trust assumptions from the identity-validation subsystem.
- Integrated multi-anchor triangulation clustering with dynamic crowd-calibrated similarity thresholds and hybrid proximity scoring for cross-subsystem, sensor-fused location validation.
- Orchestrated a roster reconciliation engine spanning absentee detection, partial-attendance classification, override workflows, audit tracking, and event-lifecycle APIs across seminars, workshops, and live monitoring dashboards.
- Hardened production infrastructure by removing hardcoded credentials, migrating to environment-secured configuration, and extending the database schema to support attendance, event, and audit reporting.

Autonomous Drone Navigation — Cognitive Flight Stack

April 2026

ROS2, PX4, AirSim, Jetson, MAVROS2 — Hybrid LLM-VLA-VLM UAV Architecture

- Led software-to-hardware integration strategy for a hybrid LLM-VLA-VLM autonomous UAV architecture, ensuring AI subsystems remained compatible with real-time flight-control constraints.
- Architected deployment pathways for perception, navigation, telemetry, and mission-planning modules across Jetson, ROS2 middleware, and PX4 flight-controller ecosystems.
- Evaluated hardware-aware tradeoffs across compute latency, sensor bandwidth, communication reliability, power consumption, and onboard resource constraints.
- Directed SITL/HITL validation workflows using PX4, AirSim, MAVROS2, and ROS2, reducing integration risk ahead of physical deployment; contributed to a three-layer cognitive flight stack spanning mission reasoning, reactive navigation, and low-level stabilization.

TECHNICAL IMPLEMENTATIONS — APPLIED ENGINEERING DEMONSTRATIONS

Satellite Change Detection System

May 2025

PyTorch, Google Earth Engine, Streamlit, Folium

- Deployed a geospatial analysis platform for land-surface change detection from Sentinel-2 imagery using a Siamese ResNet-18 pipeline for temporal feature extraction.
- Engineered real-time spatial contour segmentation with synchronized visual comparisons via Streamlit and Folium DualMap for interactive geospatial analysis.

AeroToroid

March 2025

Fusion 360, CAD, Aerodynamic Modeling

- Designed a toroidal UAV architecture with modular EDF-based propulsion and mechanically separated yaw control via a counter-rotating outer ring, eliminating standard coaxial rotor requirements.
- Produced parametric Fusion 360 models, power-distribution architecture, component selection, and cost-estimation studies for a scalable research-oriented UAV platform.

DTC Swarm Protocol

May 2026

Next.js, React Three Fiber, Supabase, PostgreSQL

- Deployed an organizational platform rendering interactive 3D drone models via React Three Fiber WebGL, with server-side session validation and role-based access workflows on Supabase/PostgreSQL.

LEADERSHIP & EXPERIENCE

Drone Technical Club (DTC), VFSTR — Founder & Lead

- Founded the campus drone-engineering student group; structured and led a 3-day hardware-build workshop mentoring 50+ students in UAV assembly and flight systems.
- Directed academic-corporate integrations, project recruitment drives, and ongoing hardware development cycles.

Campus Leadership

- Class Representative — liaison between student body and department heads for academic coordination.
- Completed Smart India Hackathon (SIH) rounds; configured localized networks and systems for regional hackathons.

CERTIFICATIONS & ACHIEVEMENTS

- Shipped two production-deployed platforms: Consensus Mesh (mobile) and DTC Swarm Protocol (web).
- HackerRank Certified — Algorithms & Data Structures
- NPTEL Certified — Business Ethics